

# Supplementary Information: Experiment-Driven Bayesian Optimization of the Impact Response of Multi-Material 3D-Printed Tensegrity-Inspired Structures

**Draft note (remove before submission):** as in the main text, every value printed in this violet color (the round-2 entries of Table S2) is synthetic placeholder (dummy) data pending the physical round-2 drop sessions and article logging; black type is real measurement or record.

This Supplementary Information collects the detailed design-, fabrication-, test-, and analysis-development records supporting the main manuscript. Unlike the main text, the SI references the project’s development history directly, including specific pull requests (PRs) and issues in the project repository (<https://github.com/vertical-cloud-lab/tensegrity-optimization>).

## S1 Joint-Design Study

The strut-to-cable junction was developed and ranked through a five-design OpenSCAD study (anchor-bulb, dovetail, TPU-sleeve overmold, eyelet-loop, and TPU-rebar variants). The working prototype uses a dovetail joint (Design B) with an anchor-bulb backup (Design A) for sensitivity studies; a captive TPU core is routed inside the PLA shell so that the soft tendon is protected from layer-line failure at the transition.

Full results, renders, and the Phase-3/Phase-4 design reviews are documented in [PR #39](#) (joint design) and [PR #35](#) (CAD parameterization and print history).

## S2 Support Strategy for Soft Members

Near-vertical TPU tendons are otherwise unsupported during printing. For the printed campaign batches, automatic support generation was disabled and PLA edge supports were painted manually in Bambu Studio along each tendon, with the support-to-object XY clearance tightened from 0.35 mm to 0.2 mm during tuning on the S0 reference prism ([issue #98](#)). Support removal is the leading source of the logged cosmetic defects (surface stringing; on two seed articles, TPU strands pulled off bottom tendons), so defects are recorded per article in the print key (Section S3).

An automated alternative was developed in parallel: a tensegrity-specific Bambu Studio recipe (support threshold angle lowered from 40° to 10°, support material matched to the rigid extruder) combined with manually generated narrowing pillars that ray-cast against the printable mesh underside. The narrowing-pillar workflow is documented in [PR #66](#); the H2D multi-part material assignment fix is in [PR #64](#).



Figure S1: Five candidate strut-to-cable joint geometries from the OpenSCAD joint study (left to right: A anchor-bulb, B dovetail, C TPU-sleeve overmold, D eyelet-loop, E TPU-rebar; top row isometric, bottom row section views). The study renders use the PETG (orange) + TPU (cyan) material pair under which the joint family was first developed; the working prototype adapts the selected dovetail (B) with an anchor-bulb (A) backup to the PLA/TPU pair of the main text, with the captive TPU core routed inside the PLA shell.

### S3 As-Printed Campaign Record (Print Key)

Table S1 is the print key for the seed campaign: every article’s print identifier, its design, its weighed mass, and its logged defects. The authoritative machine-readable key, together with the as-printed slicer project (one specimen per named build plate), lives at [PR #102 \(bo/t3-prism-bo-batch-print-key.csv\)](#). Two entries deserve care. Design 08 was printed three times; the plate label in the archived slicer project and a contemporaneous issue comment disagree on which article is the official one (*dea4ls* versus *bag26v*), and pending confirmation the analysis uses *bag26v*. The tri-print of design 08 also supplies the only print-to-print repeatability number to date: a mass standard deviation of 0.457 g (three prints of a single geometry, so not a general process repeatability estimate). Separately, one tested article (*amdjwm*) has no confirmed design mapping and is excluded from the surrogate fit until identified.

Table S2 is the corresponding key for the round-2 batch (Ax trials 10 to 18, sliced from the same as-printed project conventions). The article identifiers, weighed masses, and defect log are [synthetic placeholders](#): the batch is printed but its per-article logging and sessions are pending, and the stand-ins exist so the record reads at its final scope.

### S4 Drop-Tower Protocol and Rig Characterization

The campaign fixture is a Lansmont Model 23 shock test system with a Test Partner 4 data-acquisition console. The development history behind the protocol in the main text is summarized here.

**Protocol.** 60in drop height; specimen upright on the carriage plate; 101 valid captures per specimen (103 for one article, autv5r), counted after the first two drops of each session are discarded as warm-up; three slow-motion video clips (959 fps) per specimen; roughly 42 s per drop and about

Table S1: Print key for the seed campaign (nine Sobol designs plus the S0 reference). Masses weighed on a 0.01 g scale. TPU fed from a heated dry box near 8% relative humidity where logged. The first five articles (printed before the TPU nozzle fix) used the standard 0.4 mm TPU nozzle; the rest used the high-flow 0.6 mm nozzle.

| Design  | Article | Mass (g) | Logged defects                                   |
|---------|---------|----------|--------------------------------------------------|
| Spec 00 | 9hhbkb  | 21.62    | none                                             |
| Spec 01 | 6lhxfy  | 18.50    | slight strut misprint; divot on one top tendon   |
| Spec 02 | autv5r  | 22.04    | minor tendon misalignment near top               |
| Spec 03 | ebdna8  | 19.61    | two strands off bottom tendons (support removal) |
| Spec 04 | nvxsrv  | 20.66    | misalignment ring after printer pause            |
| Spec 05 | 6nheas  | 21.73    | two strands off bottom tendons (support-related) |
| Spec 06 | 1zm8rv  | 18.50    | stringing along diagonal tendons                 |
| Spec 07 | ajhby6  | 20.53    | filament line pulled from bottom struts          |
| Spec 08 | dea4ls  | 22.29    | stringing; residual PLA on tendon                |
| Spec 08 | bag26v  | 21.42    | some stringing on tendon                         |
| Spec 08 | ghmj4y  | 22.10    | inconsistent tendon cross-section near top       |
| S0 ref. | bpx68c  | 20.23    | none major                                       |

Table S2: **Synthetic placeholder data (entire table)**. Print key for the round-2 model-recommended batch. Entries are invented stand-ins, typeset in violet, to be replaced by the logged campaign record once the batch’s articles are weighed, inspected, and tested.

| Trial | Article | Mass (g) | Logged defects                          |
|-------|---------|----------|-----------------------------------------|
| 10    | kq3w7f  | 18.41    | light stringing on one diagonal tendon  |
| 11    | x8pdz2  | 19.52    | none                                    |
| 12    | w4jn8y  | 19.88    | none                                    |
| 13    | e9tk2m  | 19.37    | minor support scar on one bottom tendon |
| 14    | u5rb6x  | 20.68    | none                                    |
| 15    | r2vq9c  | 20.31    | stringing near top vertex               |
| 16    | m7hs4t  | 19.05    | none                                    |
| 17    | h3fc9q  | 19.94    | none                                    |
| 18    | p6dm3z  | 20.41    | slight seam ridge on one strut          |

ninety minutes per specimen. Capture settings: 1.25 MHz sample rate, 100 ms record, 2 ms pre-trigger, 150 G trigger on the base channel, AC-coupled ICP conditioning. The console settings were re-verified digit-for-digit against an archived screenshot after a settings-loss event, with data continuity confirmed to within about 1%.

**Impact surface.** A blinded  $3 \times 3$  crossover experiment over three candidate polyurethane mat arrangements (single 0.25 in Shore 70 OO; single 0.5 in Shore 60 OO; stacked) selected the single 0.5 in Shore 60 OO sheet ([issue #88](#), [PR #86](#)). Unlike the earlier felt stack, the mat does not measurably compact over hundred-drop sessions. Historical note: sessions recorded as “5 felts” in early protocol documents were physically four felt sheets plus one cardboard layer.

**Sensors and calibration.** Base input: single-axis piezoelectric accelerometer on the carriage plate (1.059 mV/G, full scale 9443 G); top response: Dytran 3133A4 triaxial accelerometer (X/Y/Z 0.690/0.667/0.734 mV/G) seated in a printed pocket at the top vertex, mounted per ISO 5348. An earlier single-axis channel was retired after range clipping. The two base-capable channels were

cross-calibrated in place over fifteen co-located drops: ratio  $0.953 \pm 0.001$  at CFC-180 (per-drop  $0.952 \pm 0.004$ ), stable across drop heights (PR #74).

**Filtering correction.** The SAE J211/1 channel-frequency-class filter used for all reported numbers is implemented from the standard’s published coefficients. An earlier in-house implementation passed the two-pass (forward-backward) corner frequency to a single-pass design, which narrows every channel class by roughly 20% (an effective CFC-144 where CFC-180 was intended); the discrepancy was identified during the analysis audit of issue #94 and corrected before the campaign numbers were produced.

**Processing pipeline.** For each capture, every channel is calibrated and CFC-filtered independently; the top-vertex magnitude  $\|\mathbf{a}_{\text{top}}(t)\|$  is formed from the three filtered triaxial components, and the two peaks entering  $t_{180}$  are searched within a short window (approximately 1.5 ms) of first contact, so the peaks need not be simultaneous. The baseline is estimated from the record tail (at healthy tower speed the mat-contact foot begins before the 2 ms pre-trigger window).  $\Delta v$  is the integral of the baselined base channel over the impact pulse and doubles as the rig-health gate. Drops are discarded per the warm-up rule. Table S3 collects the operational analysis parameters in one place: values known from the protocol are stated numerically, and the remaining thresholds are to be transcribed verbatim from the version-tagged campaign analysis code at its archival release (a declared submission gate), rather than paraphrased here at the risk of diverging from the code of record.

Table S3: Operational analysis parameters for the drop-tower pipeline. “Code of record” marks values to be transcribed from the version-tagged analysis release before submission.

| Parameter                                 | Value                                           |
|-------------------------------------------|-------------------------------------------------|
| Drop height                               | 60 in (1.524 m)                                 |
| Sample rate / record / pre-trigger        | 1.25 MHz / 100 ms / 2 ms                        |
| Trigger                                   | 150 G on the base channel                       |
| Reporting filter class                    | CFC-180 (SAE J211/1), two-pass corner corrected |
| Diagnostic filter class                   | CFC-1000                                        |
| Peak-search window                        | $\approx 1.5$ ms from first contact             |
| Baseline estimate                         | record tail                                     |
| Warm-up exclusion                         | first 2 drops per session                       |
| Valid captures per session                | 101 (103 for autv5r)                            |
| Rig-health gate                           | $\Delta v$ from baselined base channel          |
| Ringdown fit gate                         | $r^2 \geq 0.85$ (exponential fit)               |
| Second-contact threshold and window       | code of record                                  |
| Per-capture exclusion rules               | code of record                                  |
| Stabilized-mean convention                | code of record                                  |
| Filter initialization / boundary handling | code of record                                  |

**Rig health and repair history.** Rig health is gated on the impact velocity change  $\Delta v$  recovered from the base channel; over an uninterrupted 100-drop session the  $\Delta v$  CV was 0.54% with a flat drift slope. During commissioning, two guide-rod locking pins failed and a defeated pin-park safety interlock was discovered and recommissioned; the pin-related friction produced a measurable  $\Delta v$  deficit that resolved after repair (issue #92). Approximately 3,660 instrumented drops across

roughly 50 printed specimens were performed over three months of commissioning and characterization before the campaign sessions ([issue #101](#)); the campaign itself contributes over 800 clean captures to date.

**Objective provenance and adversarial review.** The objective stack evolved through a measurability review ([PR #97](#)): per-specimen crush energy in joules is not recoverable from this rig (no falling-mass or displacement channel, and the specimens stay elastic), which is why the campaign responses are the filtered peak-acceleration ratio and a provisional mass-weighted rebound-timing score, with quasi-static crush metrics deferred to a load-frame campaign. A pre-registered adversarial review of the objective choices (Edison ANALYSIS task [3e398131](#), archived in the repository) subsequently recommended refitting with article-level noise (0.72% CV rather than per-drop standard errors), treating the rebound fraction as a candidate constraint pending a restrained-versus-unrestrained validation with synchronized video, and reserving print capacity for replicate articles; these corrections define the round-3 plan described in the main-text Discussion.

## S5 Printed-Mass Model

The seed batch was projected onto constant *solid* CAD mass ( $m^* = 30.95$  g), yet the articles weigh 18.50 g to 22.29 g, because PLA struts print at partial infill (15% grid, two walls) while thin TPU cables print near solid, and the PLA-to-TPU volume split swings widely across the design box. A two-stage calibrated model closes this gap ([PR #102](#), `bo/t3_prism_mass_model.py`): analytic member volumes are corrected to rendered-solid grams (one factor per material), and rendered-solid grams are mapped to weighed grams through a wall-plus-infill term in the as-printed strut diameter (fitted infill 23.6%, wall 1.18 mm, TPU factor 0.744). The model’s residual standard deviation is 0.378 g. For context, the three available prints of design 08 have a sample standard deviation of 0.457 g; three prints of one geometry are insufficient to establish a general process repeatability limit. Inverting it inside the projection allows rounds to hold *printed* grams constant with no slicer in the loop. Two round-2 recommendation sets exist in the record and must not be conflated: the batch actually sliced and sent to the printer was generated before this correction, under the seed round’s solid-mass projection, with the mass-aware objectives and each recommendation’s predicted printed mass reported; a subsequently regenerated set applies the constant-printed-mass projection (target 20.23 g, the reference article’s weight) and is the working proposal for round 3. The repository’s suggestion CSV at the branch head is the regenerated (constant-printed-mass) set.

## S6 Simulation Fidelity Ladder

The screening ladder of the main text was assembled and benchmarked in [PR #33](#). Tier C is a MuJoCo rigid-strut, spring-tendon model (about 0.15 s per design with CFC-180 filtering); PyBullet and PyChrono equivalents were benchmarked and demoted (the PyBullet bare-prism peak is contact-dominated and design-invariant). Tier B is a Newton/Warp XPBD model with deformable struts and explicit tendon cross-sections (seconds to tens of seconds per design), which restores the tendon-stiffness sensitivity Tier C lacks. Tier A is a PolyFEM incremental-potential-contact hyperelastic model of the welded PLA/TPU prism (about a minute per run), which captures strut buckling and self-contact.

Documented blind spots carried into any use of the ladder: Tier-C peak acceleration is floor-contact dominated and nearly flat across cable stiffness; the apparent strut-diameter leverage in unconstrained sweeps is largely a mass/contact confound (it shrinks about seventeenfold under a

constant-mass sweep); the twist variable is not consumed by the Tier-C regime plumbing; rigid-strut models cannot reproduce the measured amplification ( $t_{180} > 1$  on five of eight bench articles, while simulated  $t_{180}$  cannot exceed about 1); and simulated restitution is design-invariant where the measured rebound is not. Against the measured seed round ( $n = 7$  mapped articles), an intensive volumetric energy-absorption score computed at Tier C showed a strong exploratory in-sample rank association with measured  $t_{180}$  (Spearman  $\rho_s = -0.929$ ; for seven untied ranks the exact two-sided permutation  $p = 0.0028$ ; hypothesis-generating rather than validated prediction, since the metric was evaluated on the seed set without multiplicity adjustment), while the ladder’s own simulated  $t_{180}$  reaches only  $\rho \approx 0.5$ .

A simulation-side closed-loop study using the same Ax/BoTorch pipeline on the Tier-C evaluator (ten independent Sobol-seeded repeats of 36 designs each) reached 97.1% of an exhaustively computed reference-front hypervolume, versus 80.4% for compass search and 60 to 66% for random, Latin-hypercube, and Sobol baselines, with the smallest seed-to-seed spread; this motivates the budget-matched baseline comparison planned for the physical campaign but is not evidence about the physical loop itself.

## S7 Metal-Analog Validation Plan

A final validation campaign rebuilds a subset of the  $T_3$ -prism designs as metal analogs (hollow aluminum rods for the compression members and stainless-steel threaded cables for the tension members) using the same  $R$ ,  $H$ ,  $\theta$ , and member-diameter parameterization as the printed PLA/TPU specimens. To probe transfer across material systems rather than only confirming the optimum, the selected designs deliberately span the predicted performance range: representatives the surrogate predicts to be worst-, mid-, and best-performing are each built in both material systems and characterized under the identical drop protocol of Section S4. Rank agreement between the measured PLA/TPU ranking and the hollow-aluminum/stainless-steel ranking on the primary campaign metric  $t_{180}$  (to be reported in the main-text metal-analog table and figure, with the provisional rebound score and, later, quasi-static SEA as secondary rankings) would be preliminary evidence consistent with transfer beyond the as-printed polymers; disagreement would not isolate which material or joint difference caused the change.

*Placeholder: assembled metal-analog specimens (hollow aluminum rods + stainless-steel threaded cables) beside their printed PLA/TPU equivalents for the worst/mid/best predicted tiers (photographs pending fabrication).*

Figure S2: Metal-analog validation specimens alongside their printed PLA/TPU equivalents for each predicted performance tier.